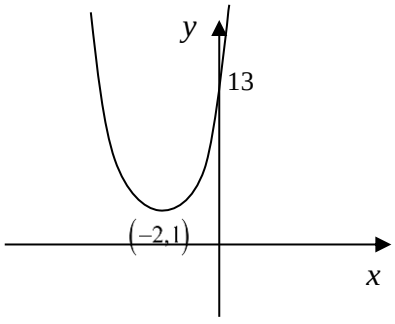
1	$4x^2-3x+7 \geqslant 4x+9$ $\Rightarrow 4x^2-7x-2\ldots 0 \Rightarrow (4x+1)(x-2)\ldots 0 \Rightarrow x=\ldots$ or $\Rightarrow 4x^2-7x-2\ldots 0 \Rightarrow x=\dfrac{-(-7)\pm\sqrt{(-7)^2-4(4)(-2)}}{2\times 4} \Rightarrow x=\ldots$ or $\Rightarrow 4x^2-7x-2\ldots 0 \Rightarrow 4\left(x^2-\dfrac{7}{4}x-\dfrac{1}{2}\right)\ldots 0 \Rightarrow 4\left(\left(x-\dfrac{7}{8}\right)^2-\left(\dfrac{7}{8}\right)^2-\dfrac{1}{2}\right)\ldots 0 \Rightarrow x=\ldots$	**M1**
	$x=-\dfrac{1}{4}, 2$	**A1**
	$x \leqslant "-\dfrac{1}{4}", \quad x \geqslant "2"$	**M1**
	$x \leqslant -\dfrac{1}{4}$ or $x \geqslant 2$ oe	**A1**
		(4)

2(a)	$2x+2y=350$	**B1**
		(1)
(b)	E.g. $xy=7350$, $x\times y=7350$	**B1**
		(1)
(c)	$x(175-x)=7350$ or $(175-y)y=7350$	**M1**
	E.g. $x^2-175x+7350=0\Rightarrow\left(x-70\right)\left(x-105\right)=0\Rightarrow x=...$	**dM1**
	x = 70 or 105	**A1**
	$(x>y\Rightarrow)x=105, y=70$	**A1**
		(4)

3(a)	$a = 3$		**B1**
	$b = \pm 2$		**M1**
	$3x^2 + 12x + 13 = 3\left(x + 2\right)^2 + 1$ or $a = 3,\ b = 2,\ c = 1$		**A1**
			(3)
(b)		Correct U shape with minimum in second quadrant	**B1**
		Intercept 13 on y-axis.	**B1**
		Vertex at $\left(-2,1\right)$	**B1ft**
			(3)

4(a)	$y = \dfrac{5x^2 + \sqrt{x^3}}{\sqrt[3]{8x}} = \dfrac{5x^2 + x^{\frac{3}{2}}}{...x^{\frac{1}{3}}}$	**M1**
	$= \dfrac{5x^{2-\frac{1}{3}}}{2} + \dfrac{x^{\frac{3}{2}-\frac{1}{3}}}{2}$	**M1**
	$= \dfrac{5}{2}x^{\frac{5}{3}} + \dfrac{1}{2}x^{\frac{7}{6}}$	**A1A1**
		(4)
(b)	$\left(\dfrac{\mathrm{d}y}{\mathrm{d}x} =\right) ...x^{\frac{2}{3}} + ...x^{\frac{1}{6}}$	**M1**
	$\left(\dfrac{\mathrm{d}y}{\mathrm{d}x} =\right) \dfrac{25}{6}x^{\frac{2}{3}} + \dfrac{7}{12}x^{\frac{1}{6}}$	**A1ftA1**
		(3)

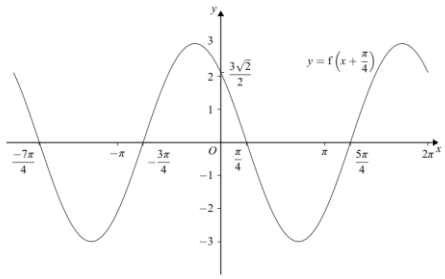
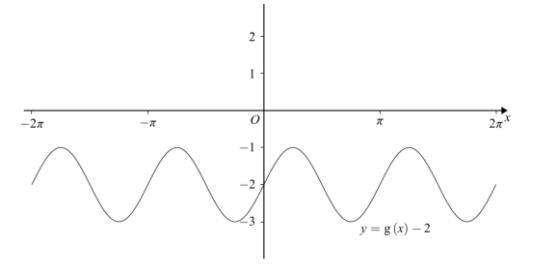
5(a)	$\angle AOB = \cos^{-1}\frac{1.5}{4} = ...$ or $\angle AOB = \tan^{-1}\frac{\sqrt{4^2-1.5^2}}{1.5} = ...$ or $\angle AOB = \frac{\pi}{2} - \sin^{-1}\frac{1.5}{4} = \ldots$ or $\angle AOB = \frac{\pi}{2} - \tan^{-1}\frac{1.5}{\sqrt{4^2-1.5^2}} = ...$	**M1**
	= 1.186 *	**A1***
		(2)
(b)	$l = 4\times 1.186\left(= 4.744\right)$	**M1**
	$BC = 6 - 4\sin 1.186\left(= 6 - 3.708... = 2.29...\right)$ or $BC = 6 - 4\cos\left(\frac{\pi}{2} - 1.186\right)\left(= 6 - 3.708... = 2.29...\right)$ or $BC = 6 - \sqrt{4^2-1.5^2}\left(= 6 - 3.708... = 2.29...\right)\left(= \frac{12-\sqrt{55}}{2}\right)$ or $BC = 6 - \frac{1.5}{\tan"0.384"}\left(= 6 - 3.708... = 2.29...\right)$	**M1**
	Perimeter is $4\times 1.186... + \left(6 - 4\cos 0.384...\right) + 1.5 + 6 + 4 = ..$	**ddM1**
	= awrt 18.5 (m)	**A1**
		(4)
(c)	Area sector = $\frac{1}{2}\times 4^2 \times 1.186...$	**M1**
	Area *OBCD* Examples: $6\times 1.5 - \frac{1}{2}\times"3.708"\times 1.5$ or $\left(6-"3.708"\right)\times 1.5 + \frac{1}{2}\times"3.708"\times 1.5$ or $\frac{1}{2}\left(6+6-"3.708"\right)\times 1.5$ or $6\times 1.5 - \frac{1}{2}\times 1.5\times 4\sin 1.186$ or $\frac{1}{2}\times 6\times 1.5 + \frac{1}{2}\left(6-"3.708"\right)\times 1.5$ (= 6.219…)	**M1** **A1 on EPEN**
	$\frac{1}{2}\times 4^2 \times 1.186 + 6.219... =$	**ddM1**
	= awrt 15.7 (m^2)	**A1**
		(4)

6(a)	$\left(r-\frac{1}{r}\right)^2 = r^2 - r\times\frac{1}{r} - r\times\frac{1}{r} + \frac{1}{r^2}$	**M1**
	$= r^2 + \frac{1}{r^2} - 2$	**A1**
		(2)
(b)	$\frac{1}{3+2\sqrt{2}} = \frac{1}{3+2\sqrt{2}}\times\frac{3-2\sqrt{2}}{3-2\sqrt{2}}$	**M1**
	$= \frac{3-2\sqrt{2}}{3^2-\left(2\sqrt{2}\right)^2} = 3-2\sqrt{2}$	**A1**
		(2)
(b) ALT	$\frac{1}{3+2\sqrt{2}} = p+q\sqrt{2} \Rightarrow 1 = \left(p+q\sqrt{2}\right)\left(3+2\sqrt{2}\right) = 3p+4q+2p\sqrt{2}+3q\sqrt{2}$ $\Rightarrow \begin{matrix} 3p+4q=1 \\ 2p+3q=0 \end{matrix}$	**M1**
	$\Rightarrow \begin{matrix} 3p+4q=1 \\ 2p+3q=0 \end{matrix} \Rightarrow \begin{matrix} 6p+8q=2 \\ 6p+9q=0 \end{matrix} \Rightarrow q=-2 \Rightarrow 3p-8=1 \Rightarrow p=3$ $\frac{1}{3+2\sqrt{2}} = 3-2\sqrt{2}$	**A1**
		(2)
(c)	$\left(\sqrt{3+2\sqrt{2}} - \frac{1}{\sqrt{3+2\sqrt{2}}}\right)^2 = 3+2\sqrt{2} + \frac{1}{3+2\sqrt{2}} - 2$	**M1**
	$= 3+2\sqrt{2}+3-2\sqrt{2}-2 = ...(=4)$	**dM1**
	so $\sqrt{3+2\sqrt{2}} - \frac{1}{\sqrt{3+2\sqrt{2}}} = 2$	**A1**
		(3)
(c) Alt	$\sqrt{3+2\sqrt{2}} - \frac{1}{\sqrt{3+2\sqrt{2}}} = 2 \Rightarrow 3+2\sqrt{2}-1 = 2\sqrt{3+2\sqrt{2}}$	**M1**
	$\Rightarrow \left(2+2\sqrt{2}\right)^2 = 4\left(3+2\sqrt{2}\right)$	**dM1**
	$\Rightarrow 4+8\sqrt{2}+8 = 12+8\sqrt{2}$ ✓ Hence true	**A1**
		(3)

7(a)	$0=10-2x \Rightarrow x=5$ or $y=2, y=10-2x \Rightarrow x=4$	**B1**
	Examples: $\frac{1}{2}\times 2\left(5+4-a\right)=\frac{27}{4}$ or $\frac{1}{2}\times 2\left(5+4-\frac{2}{k}\right)=\frac{27}{4}$ Trapezium or $\frac{1}{2}\times 2a+\frac{1}{2}\times 2\left(5-a+4-a\right)=\frac{27}{4}$ or $\frac{1}{2}\times 2\times\frac{2}{k}+\frac{1}{2}\times 2\left(5-\frac{2}{k}+4-\frac{2}{k}\right)=\frac{27}{4}$ Triangle + Trapezium or $\frac{1}{2}\times 2a+2\left(4-a\right)+\frac{1}{2}\times 1\times 2=\frac{27}{4}$ or $\frac{1}{2}\times 2\times\frac{2}{k}+2\left(4-\frac{2}{k}\right)+\frac{1}{2}\times 1\times 2=\frac{27}{4}$ Triangle + Rectangle + Triangle or $\frac{1}{2}\times 5\times 2+\frac{1}{2}\left(4-a\right)\times 2=\frac{27}{4}$ or $\frac{1}{2}\times 5\times 2+\frac{1}{2}\left(4-\frac{2}{k}\right)\times 2=\frac{27}{4}$ 2 Triangles	**M1**
	$\Rightarrow k=\frac{8}{9}, a=\frac{9}{4}$	**A1** **A1ft**
		(4)
(b)	Two of $y \geqslant "\frac{8}{9}" x,\ y \leqslant 10-2x, x > "\frac{9}{4}"$	**M1**
	All three of $y \geqslant \frac{8}{9}x,\ y \leqslant 10-2x, x > \frac{9}{4}$	**A1**
		(2)

8(a)	$y=\frac{1}{4}x^3-8x^{-\frac{1}{2}} \Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right)\frac{1}{4}\times 3x^2-8\times-\frac{1}{2}x^{-\frac{3}{2}}$	**M1 A1**
	$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right\|_{x=4}=\ldots\left(\frac{25}{2}\right)$	**M1**
	$y-12=\text{"}\frac{25}{2}\text{"}\left(x-4\right)$	**dM1**
	$25x-2y-76=0$ oe e.g. $-25x+2y+76=0$	**A1**
		(5)
(b)	$\left(\mathrm{f}(x)=\right)\int\frac{1}{4}x^3-8x^{-\frac{1}{2}}\,\mathrm{d}x=\frac{1}{4}\frac{x^4}{4}-\frac{8x^{\frac{1}{2}}}{\frac{1}{2}}(+c)$	**M1 A1**
	$\mathrm{f}(4)=12\Rightarrow 16-32+c=12\Rightarrow c=\ldots(28)$	**dM1 A1**
	So $\left(\mathrm{f}(x)=\right)\frac{x^4}{16}-16\sqrt{x}+\text{"28"}$	**A1ft**
		(5)

8(a)	$y=\frac{1}{4}x^3-8x^{\frac{1}{2}} \Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right)\frac{1}{4}\times 3x^2-8\times\frac{1}{2}x^{-\frac{1}{2}}$	**M1 A1**
	$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right\|_{x=4}=\ldots(10)$	**M1**
	$y-12="10"\left(x-4\right)$	**dM1**
	$10x-y-28=0$ oe e.g. $-10x+y+28=0$	**A1**
		(5)
(b)	$\left(\mathrm{f}(x)=\right)\int\frac{1}{4}x^3-8x^{\frac{1}{2}}\,\mathrm{d}x=\frac{1}{4}\frac{x^4}{4}-\frac{8x^{\frac{3}{2}}}{\frac{3}{2}}(+c)$	**M1 A1**
	$\mathrm{f}(4)=12\Rightarrow 16-\frac{128}{3}+c=12\Rightarrow c=\ldots\left(\frac{116}{3}\right)$	**dM1 A1**
	So $\left(\mathrm{f}(x)=\right)\frac{x^4}{16}-\frac{16}{3}x^{\frac{3}{2}}+"\frac{116}{3}"$	**A1ft**
		(5)

9(i)(a)	$\left(y=\right)3\cos\left(x\right)$		**M1 A1**
			(2)
(b)		Same shape translated left or right	**B1**
		All x intercepts labelled correctly.	**B1**
		Correct y intercept $\frac{3\sqrt{2}}{2}$	**B1**
			(3)
(ii)(a)	$\left(y=\right)\sin(2x)$		**M1 A1**
			(2)
(b)		Same shape translated down below the x-axis.	**B1**
		Correct y intercept -2 labelled.	**B1**
			(2)

10(a)(i)	Equation is $y=\frac{1}{2}\left(x+2\right)$	**B1**
		(1)
(ii)	l_1 intersect parabola $\Rightarrow -\frac{1}{4}\left(x+2\right)\left(x-b\right)=\frac{1}{2}\left(x+2\right)\Rightarrow x=...$	**M1**
	$x=b-2$	**A1**
	$y=\frac{1}{2}b$	**A1**
		(3)
(b)	l_2 passes through $(b, 0)$ and has gradient -2 $\Rightarrow y=...$	**M1**
	$y-0=-2\left(x-b\right)$	**A1**
		(2)
(c)	So equation is $y-"\frac{1}{2}b"="-2"\left(x-"b-2"\right)$	**M1**
	$y-\frac{1}{2}b=-2x+2b-4\Rightarrow y=-2x+\frac{5}{2}b-4$ *	**A1***
		(2)
(d)	$y=-2x+2b=-2x+\frac{5}{2}b-4\Rightarrow 2b=\frac{5}{2}b-4\Rightarrow b=...$ or $y=-2x+\frac{5}{2}b-4,\ x=b,\ y=0\Rightarrow 0=-2b+\frac{5}{2}b-4\Rightarrow b=...$ or $-\frac{1}{4}\left(x+2\right)\left(x-b\right)=-2\left(x-b\right)\Rightarrow x=...\left(6\right)\Rightarrow b-2=6\Rightarrow b=...$	**M1**
	$b = 8$	**A1**
		(2)